Change the mass and watch the acceleration

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Fair test / ES: prueba controlada / PT: teste controlado

Change one thing at a time.

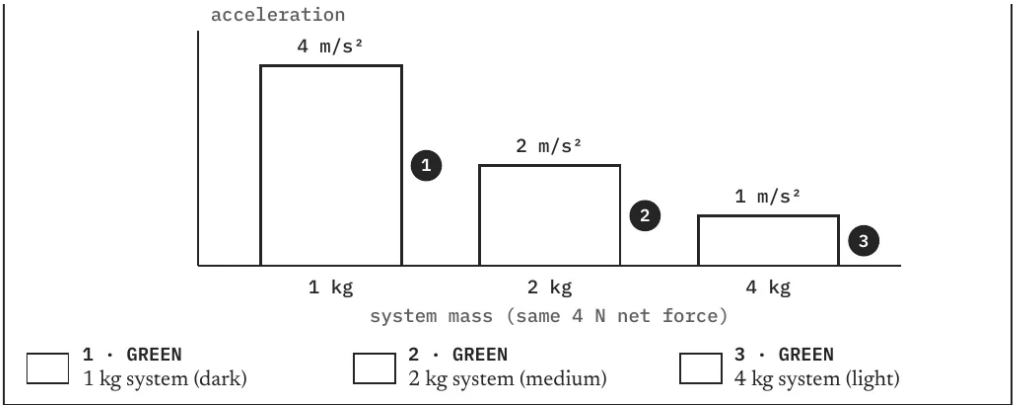
Also on your note page: Changed variable, Controls, Evidence.

B READ THE RELATIONSHIP

To test how mass matters, change mass and keep the net force the same.

At fixed net force, more mass means less acceleration

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

In one trial, velocity rises from 0 to 4 m/s in 2 s. Find the acceleration.

Given $v_i = 0$ m/s $v_f = 4$ m/s $\Delta t = 2$ s

Rule $a = \Delta v \div \Delta t$

Work $\Delta v = 4 - 0 = 4$ m/s $a = 4 \div 2$

Answer 2 m/s² use your own measured graph

E SAVE FOR THE FINAL MEETING

A massive object is hard to nudge. The bigger XJ's mass, the smaller the acceleration any push or pull gives it, which matters for any plan to deflect it.

Read the physics: Fair test

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Compare a light cart with a loaded cart. Use the same net force and the same release method. Otherwise a change in acceleration could have more than one cause.

02 HOW IT WORKS

Read acceleration from the slope of a velocity–time graph. At fixed net force, doubling the moving mass halves acceleration. Multiplying mass by acceleration gives the same net force in both trials.

03 WHAT TO CHECK

Keep the hanging mass, track and release conditions consistent. Name what changed and what was controlled before stating the result.

04 TRY THE EXAMPLE FROM YOUR NOTES

In one trial, velocity rises from 0 to 4 m/s in 2 s. Find the acceleration. 2 m/s² use your own measured graph.

05 CONNECT TO XJ

A massive object is hard to nudge. The bigger XJ's mass, the smaller the acceleration any push or pull gives it, which matters for any plan to deflect it.

SAY IT BEFORE YOU WRITE

Which quantity must match in the two trials? We changed ___. We kept ___ fixed. Acceleration ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Fair test / ES: prueba controlada / PT: teste controlado

Say it in any language. Draw or use a few words. Use the notes to check your idea.

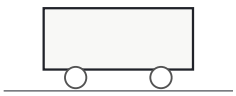
Meaning	How can you tell?
To me, fair test means …	To test the effect of mass, what will you change? What will you keep the same? I will change ___ and keep ___ the same.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

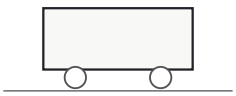
Draw two masses with the same force. Label what changes, what stays fixed, and which accelerates more.

Labels you can use: change · control · compare.

Lower mass



Higher mass



Tell a partner: We changed ___. We kept ___ fixed. Acceleration ___.

On assignment: Fair test

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two asteroid models get the same net force for the same time. Model H has twice the mass of L.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Test engineer**

Build a two-row test card. Pick the model with the larger velocity change and show the mass comparison for the test crew.

☐ **Lab correspondent**

Explain to viewers why two equal pushes can look different. Your producer must check what stayed the same in both trials.

Word bank: change · control · compare.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”